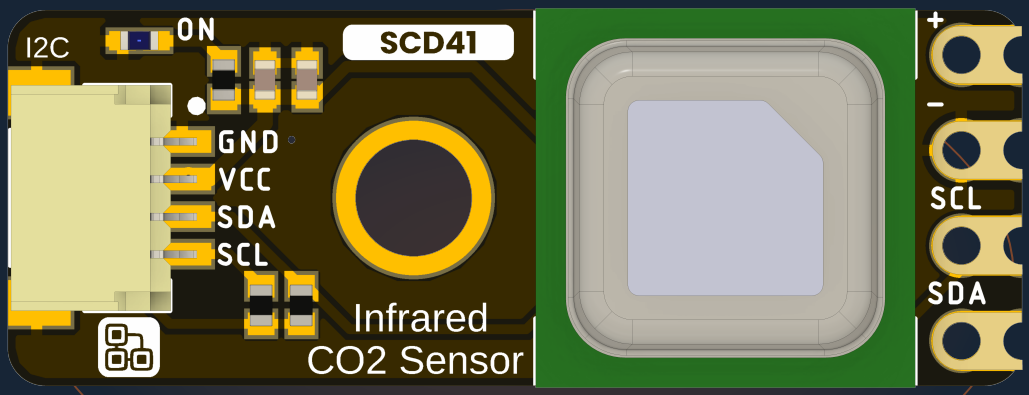
PRODUCT REFERENCE MANUAL

UNIT ELECTRONICS · DEVLAB ECOSYSTEM

DevLab I2C SCD41 Air Quality Sensor

MPN: UE0179

Version 1.1.0



I2C air-quality sensor module based on the SCD41.

AIR QUALITY | I2C | DEVLAB ECOSYSTEM

DOCUMENT MAP

Contents

Description
Applications
Available Documentation

1 The Board
1.1 Board Identification
1.2 Board Views

2 Ratings
2.1 Electrical Limits
2.2 Interface Characteristics

3 Functional Overview
3.1 Integration Boundary

4 Connectors & Pinouts
4.1 Released Pinout
4.2 Connection Groups

5 Board Operation
5.1 Before Connecting
5.2 I2C Bus Check

6 Mechanical Information
6.1 Top View
6.2 Bottom View
6.3 Board Dimensions

7 Company Information

8 Reference Documentation

9 Appendix
9.1 Schematic Source
9.2 Document Control
9.3 Data Requiring Confirmation
9.4 Board Schematic

Description

The DevLab I2C SCD41 Air Quality Sensor (MPN **UE0179**) is an I2C air quality sensing module in the UNIT Electronics DevLab family. The product-specific board views and any released drawings are collected under `hardware/`.

Applications

- Air quality sensing in an I2C-based prototype.
- Evaluation of the module with a compatible host controller.

Available Documentation

The UE0179 board schematic and pinout appear under `hardware/`. Chapter 8 lists the source files, and Appendix 9.4 shows the complete schematic.

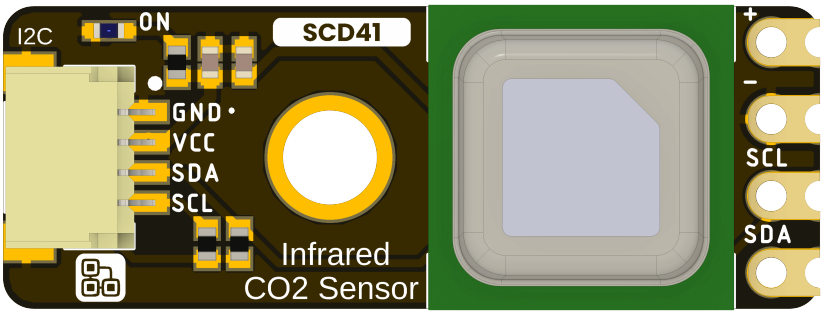
1 The Board

1.1 Board Identification

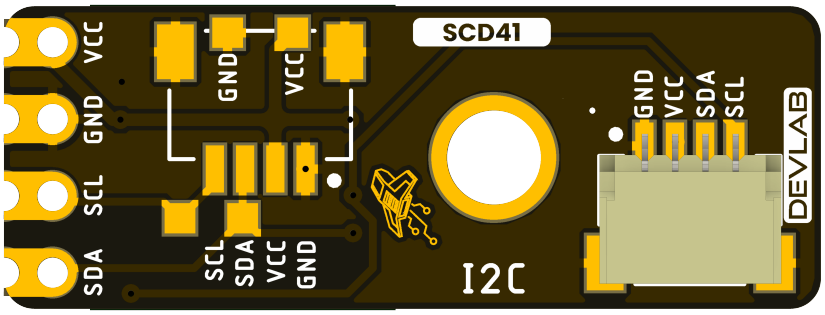
Item	Identification
Product	DevLab I2C SCD41 Air Quality Sensor
Manufacturer Part Number (MPN)	UE0179
Function	Air quality sensing

1.2 Board Views

The following views come from the selected UE0179 source package. They show component placement and board orientation; use a dimensioned drawing for mechanical tolerances.



DevLab I2C SCD41 Air Quality Sensor top view.



DevLab I2C SCD41 Air Quality Sensor bottom view.

2 Ratings

The selected UE0179 source package does not establish a complete set of electrical ratings in this reference. Do not use values from another DevLab module as substitutes.

2.1 Electrical Limits

Supply range, I/O voltage limits, and current consumption require confirmation from the product-specific schematic and component data before integration.

2.2 Interface Characteristics

Confirm the I2C address, bus speed, pull-up arrangement, and connector orientation with the released UE0179 pinout or firmware documentation. An I2C scan only identifies responding addresses; it does not verify the device protocol.

3 Functional Overview

The DevLab I2C SCD41 Air Quality Sensor provides air quality sensing for an I2C-based system. The UE0179 schematic shows the SCD41 sensor connected to VSYS, GND, SDA, and SCL. It shows SDA and SCL pull-ups to VSYS, supply decoupling capacitors, a power indicator, and three I2C hub connectors.

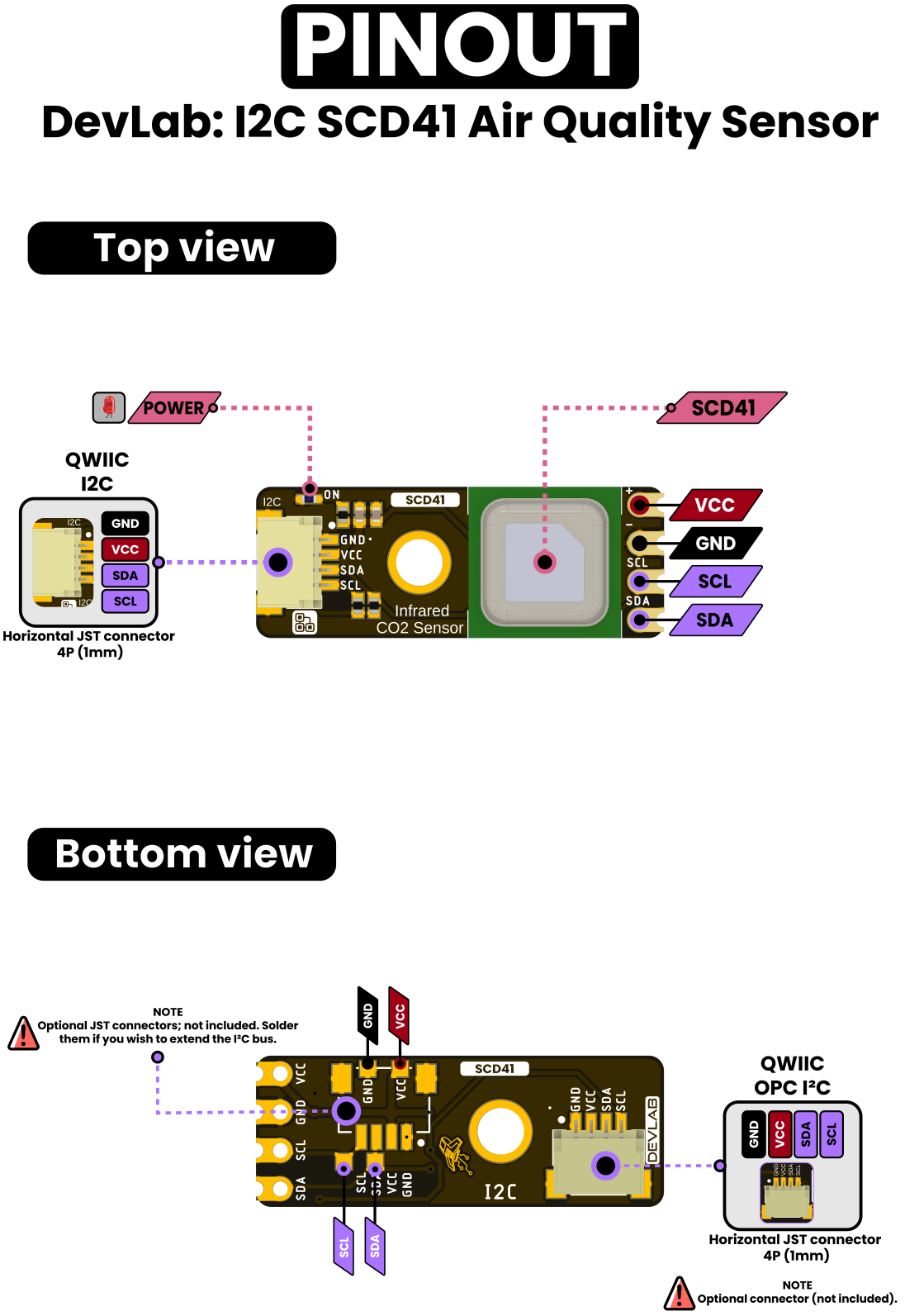
3.1 Integration Boundary

The UE0179 pinout and schematic identify the connector signals and board connections. The firmware release and sensor documentation establish the I2C address and commands. The generic I2C scanner in `software/` serves as a bus diagnostic.

4 Connectors & Pinouts

The released V1.0 pinout for the UE0179 SCD41 Air Quality Sensor shows the top and bottom board views. Check the view and connector orientation before wiring.

4.1 Released Pinout



UE0179 SCD41 Air Quality Sensor pinout, top and bottom views.

Figure 4.1 — UE0179 pinout, English version. The Spanish version is linked in Section 8.

4.2 Connection Groups

Group	Labels in the released drawing	Purpose
Qwiic I2C connector	GND, VCC, SDA, SCL	Power and I2C bus connections
Board contacts	VCC, GND, SCL, SDA	Direct access to the labeled power and bus signals

The drawing also marks optional 4-pin, 1 mm JST connector positions. The pinout does not establish electrical limits or connector part numbers.

5 Board Operation

5.1 Before Connecting

Check the product-specific UE0179 pinout or schematic for power, ground, SDA, and SCL. The board views alone do not establish voltage limits or connector numbering.

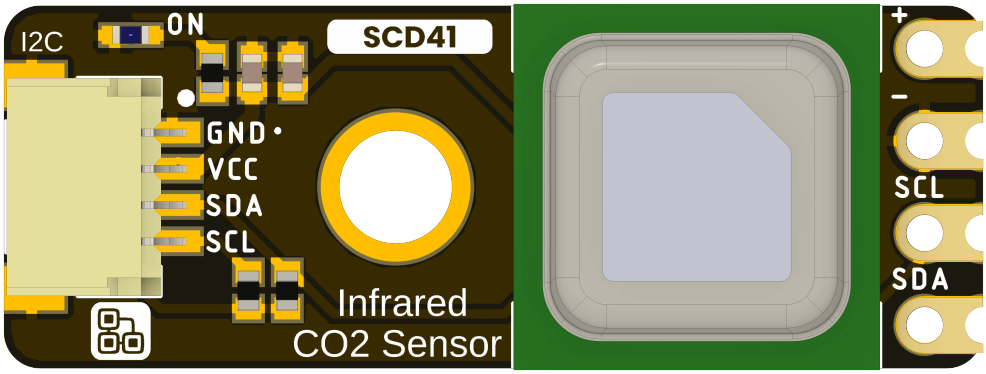
5.2 I2C Bus Check

After confirming the wiring and supply limits, use the generic Arduino or MicroPython I2C scanner in `software/examples/i2c/`. Set the host SDA and SCL pins for your controller. A responding address confirms only that a device acknowledged on the bus; it does not identify the module or validate its measurement protocol.

6 Mechanical Information

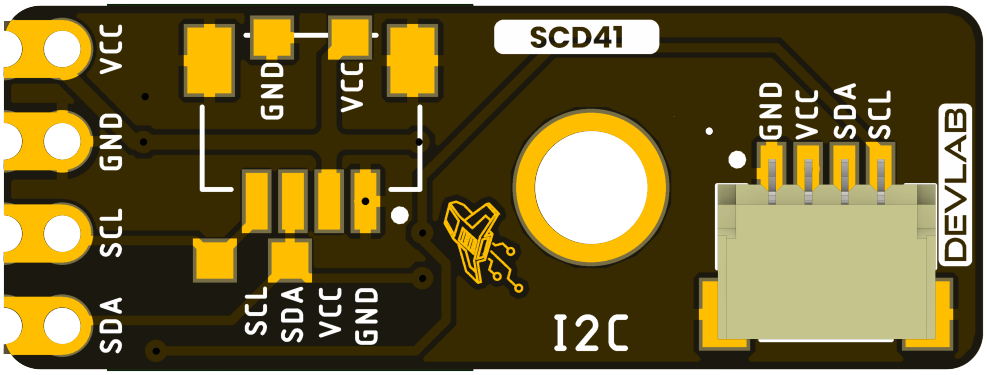
The supplied board views identify the component and underside layouts of the UE0179 I2C SCD41 Air Quality Sensor. Use the labeled drawings for fit checks; do not infer dimensions by scaling a render.

6.1 Top View



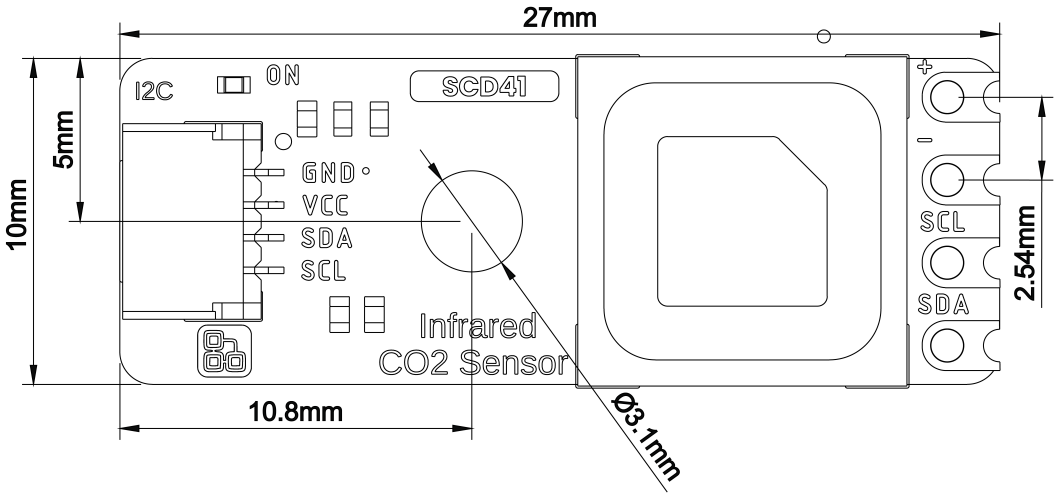
UE0179 I2C SCD41 Air Quality Sensor top view.

6.2 Bottom View



UE0179 I2C SCD41 Air Quality Sensor bottom view.

6.3 Board Dimensions



Board Dimensions

UE0179 I2C SCD41 Air Quality Sensor dimension drawing.

Use the dimensions printed on the supplied drawing. PCB thickness, component heights, connector clearance, and manufacturing tolerances require separate verification.

7 Company Information

Item	Information
Company / brand	UNIT Electronics
Board ecosystem	DevLab
Product family	Atom
Website	UNIT Electronics
Address	Salvador 19, Cuauhtémoc, 06000 Mexico City, CDMX, Mexico

The product repository and controlled technical references are listed in the next chapter.

8 Reference Documentation

Reference	Location
Product repository	UE0179 SCD41 Air Quality Sensor
UE0179 board schematic	PDF
Pinout V1.0, English	PDF · PNG
Pinout V1.0, Spanish	PDF · PNG
Top view	3D board view
Bottom view	3D board view
Board dimensions	Dimension drawing
Board topology	Topology drawing
CAD model	STEP

The released pinout documents the visible labels and connector orientation. It does not by itself establish supply limits, bus timing, or mating part numbers.

9 Appendix

9.1 Schematic Source

The UE0179 V1.0 board schematic identifies the SCD41 sensor, I2C connectors, pull-ups, supply decoupling, power indicator, and test points.

9.2 Document Control

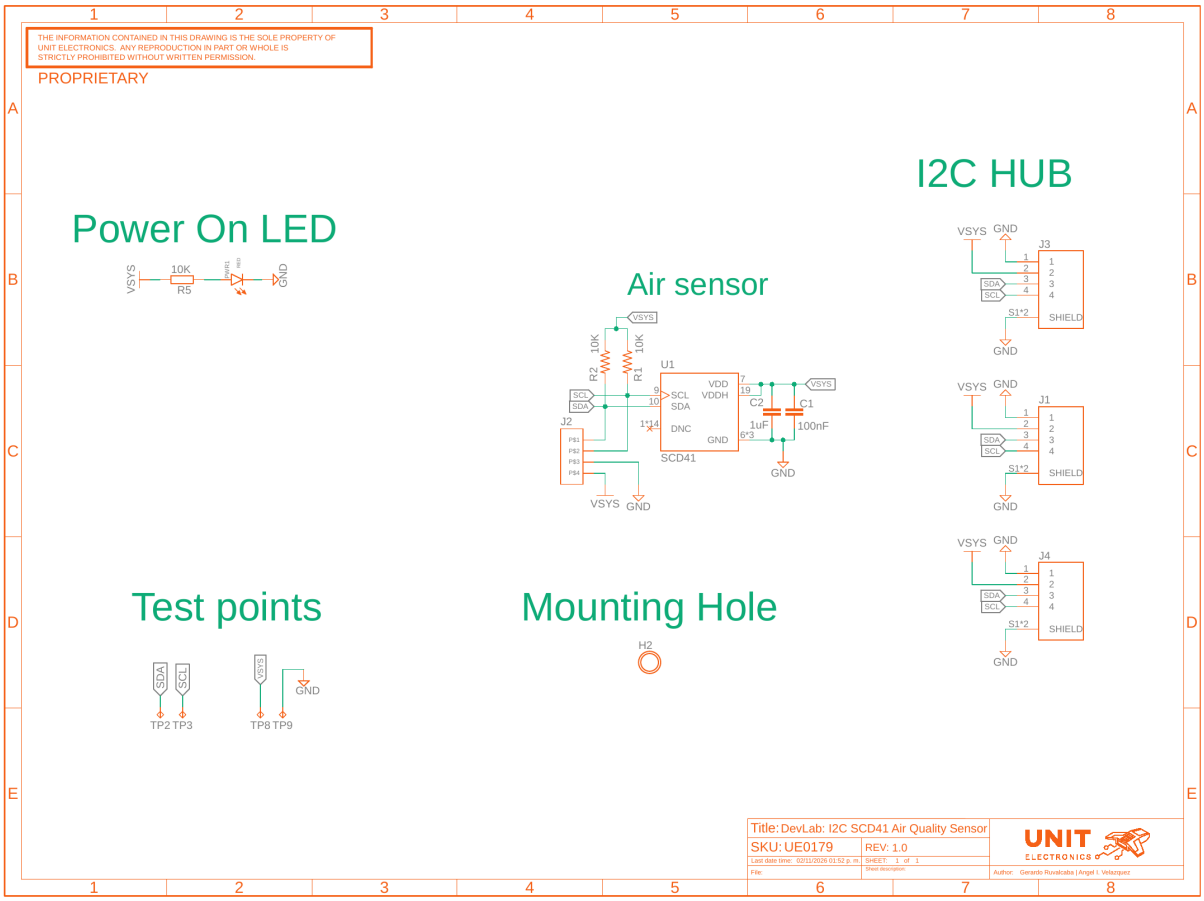
Item	Value
Product	DevLab I2C SCD41 Air Quality Sensor
MPN	UE0179
Source	UNIT Electronics selected product source package

9.3 Data Requiring Confirmation

Electrical limits, I2C commands, and mechanical tolerances must be verified against product-specific released documentation.

9.4 Board Schematic

The UE0179 board schematic shows the complete sensor and connector circuit.



UE0179 I2C SCD41 Air Quality Sensor board schematic.